Matched Preference Labels Under Source Failures

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Abstract

Matched preference labels should agree or reverse under specified changes to a scenario. Signed-graph balance detects inconsistent rules and gives a cycle that an editor can inspect. When reference labels may share a faulty source, a vertex-cover calculation identifies which label orientations are still forced. Exhaustive finite checks verify both reductions. A separate lexical experiment on public deontology annotations compares responses within the same scenario, tests the hard split, and measures how often the same responses change ranking across scenarios. The structural guarantee requires valid matching rules and a source-fault budget.

Summary and scope

Question. What can paired supervision establish when comparisons are consistent but the reference labels are uncertain?

Finding. A signed graph certifies consistency. A separate vertex-cover test certifies which orientations remain forced under a stated source-fault budget.

Evidence and scope. The reductions are proved and checked with exact oracles. A fitted lexical study on a public corpus evaluates matching as a diagnostic, without attesting the correctness of an annotation, a person's preferences, or an agent's future actions.

1 What a swap should preserve

Counterfactual fairness treats certain demographic changes as interventions whose effect on a decision depends on a causal model [1]. During data construction the narrower task is to enforce a declared symmetry only when relevant roles, permissions and facts are held fixed. Swapping a name under unchanged authority may leave the preferred action the same. Swapping the roles of the person who gave consent and the person who did not can legitimately change it. A pair needs a reason for its sign.

Label a finite set of matched cases by $y_i \in \{-1, 1\}$, meaning one of two clearly ordered responses wins. Each edge requests $y_i y_j = s_{ij}$, where sign $+1$ preserves the preference and sign -1 reverses it. The graph-balance characterization is the classical criterion of Harary, as presented by Zaslavsky [2]. For example, if two successive swaps preserve a label, the direct comparison between their endpoints cannot reverse it. The three edges form a negative cycle. An editor can inspect those three transformation rules before deciding whether to collect more labels.

Proposition 1. *The constraints are satisfiable exactly when every cycle has positive sign product.*

Proof. Multiplying the constraints around a cycle cancels each y_i twice, giving one. Conversely fix a root label in each component and propagate labels by multiplying along a spanning tree. Every non-tree edge closes a cycle. Its sign equals the product along the tree path by positivity of that cycle, so its constraint is satisfied. □

2 What the developer should do with a conflict

When a cycle fails, preserve the source cases and the transformation that generated each edge. Review the first inconsistent cycle before tuning the model. A missing role distinction can justify deleting or reversing an edge with a logged reason. Ambiguous consent may need a third response type, which lies outside the binary model. Anchor valid components with independently adjudicated examples. Without an anchor, flipping every label in a component preserves all edges, so structural consistency cannot identify which action is actually appropriate.

3 A shared fault budget for anchor sources

After the balance check, fix one satisfying labeling g_i on each component. Every other satisfying labeling has the form $y_i = x_{C(i)} g_i$, where $x_c \in \{-1, 1\}$ is the orientation of component c. An anchor from root r asserting $y_i = b$ endorses $x_{C(i)} = b g_i$. All reports traced to the same original adjudication are charged to the same root, even if they arrive through different reviewers. Assume the signed relations are correct and that some set of at most f roots accounts for every incorrect anchor. No probabilistic independence assumption is needed.

Let F contain the roots that endorse both signs of any one component. Form the conflict graph H on the other roots, joining two roots when their endorsements disagree on a component. For a partial orientation z, let F_z be F together with every root that contradicts any of its pinned orientations. Write $\tau(G)$ for the minimum size of a vertex cover of G.

Proposition 2. *A partial orientation z extends to a labeling compatible with the fault budget exactly when*

$$|F_z| + \tau(H - F_z) \leq f.$$

Proof. Every root in F_z must be removed. Every edge left in $H - F_z$ has incompatible endpoints, so the remaining removals must cover all its edges. Conversely, remove F_z and a cover of those edges. Each component now receives either one sign or no sign from the surviving roots. Its sign can be set to that unanimous endorsement, to its pin if it has one, or freely in the absence of both. The resulting labeling satisfies all the signed pairs and all surviving anchors. □

Test both pins for each component. If only one is feasible, the budget forces it. If both are feasible, retain the ambiguity. If neither is feasible, the budget and reports are jointly inconsistent. For the bounded cover search, choose a remaining edge and branch on removing either endpoint. A residual budget of k therefore requires at most $2^{k+1} - 1$ search nodes, with polynomial work at each node. For unbounded f the feasibility problem is NP-complete by the classical vertex cover problem [3]. To realize an arbitrary conflict graph, introduce one singleton component per edge and have the two endpoint roots endorse opposite signs on it. A set of removed roots is a certificate whose consistency is checked directly.

Suppose root A endorses the positive sign on two components, while roots B and C each endorse the negative sign on a different one. A budget of one leaves either sign possible when each component is examined alone. Jointly, the only one-root cover is {A}, and both orientations are negative. If

A's reports are copied through more reviewers, the conclusion is unchanged when provenance is preserved. An editor can use this test to direct adjudication toward components that remain ambiguous. If the signed comparisons, source identities, or fault budget are wrong, the result does not certify the resulting labels.

4 Exhaustive check and model experiment

Each pair of four vertices can be absent, positive, or negative. The audit compares traversal with all sixteen complete assignments for $3^6 = 729$ signed graphs. Both methods classify 393 as consistent. The three-edge witness with two positive and one negative edge has no solution. For the source-fault extension, direct orientation enumeration agreed with the cover solver on 472,392 pinned queries across 19,683 ternary matrices and four budgets. In 9,792 component-and-budget cases a feasible global model forced a sign that the isolated component left ambiguous. The solver also agreed with exhaustive source-subset search on all 1,024 five-root conflict graphs, with six budgets per graph. These counts describe the finite test design. A later study should inject matched-example contradictions and compare training after source-level review with training after discarding examples without explanation. Evaluate justified asymmetries separately from identity invariance. Consistency of declared pairs still depends on their causal and normative validity.

5 Learning from independently anchored pairs

We generated paired supervision with an unchanged task signal and a swapped identity cue. In a designated quadrant we corrupted 90% of positive labels. The repaired training arm used the simulator's independent truth for each pair as its anchor and propagated the known preserving sign to the other member. Both arms fitted logistic scorers of the same form. On 3,200 held-out pairs, the raw labels produced 1,646 swap disagreements and 4,740 correct individual decisions out of 6,400. Anchored labels yielded zero disagreements and 6,400 correct decisions. This experiment models valid anchor information. Agreement alone would also allow both members of a pair to be wrong.

6 Matched scoring on public deontology judgments

The ETHICS deontology files pair scenarios with responses and include binary judgments [4]. They provide multiple responses to the same scenario. This experiment asks how well a lexical scorer orders those responses. It does not infer signed invariances or adjudicate the provenance of individual labels.

Before fitting, we fixed four feature views and hashed the data. We fit add-one lexical likelihood ratios from 18,161 training rows after removing three duplicates. Scenario words, response words and a small relation block have separate likelihood ratios. Each vocabulary keeps features seen at least three times. The relation block includes shared tokens, a capped overlap count and joint negation flags. Compare scenario alone, response alone, their additive sum, and the additive sum with the relation block. The same training class prior is used for every row classification decision. There is no test tuning.

We exclude test scenarios whose case-and-whitespace-normalized text appears in training. This removes 1,300 ordinary rows and 736 hard rows. The remaining splits share one normalized scenario with each other. This filter addresses exact scenario overlap and does not test for paraphrases. Within a scenario, compare every label-one response with every label-zero response, award half

credit for ties, and average the resulting proportions equally across scenarios. There are 2,252 comparisons across 574 ordinary scenarios and 2,742 across 700 hard scenarios.

	Ordinary, 2,296 rows		Hard, 2,800 rows	
Score	Row correct	Pair order	Row correct	Pair order
Scenario	1,160	0.5000	1,402	0.5000
Response	1,618	0.8077	1,696	0.6597
Additive	1,580	0.8077	1,693	0.6597
Additive with relations	1,569	0.8046	1,693	0.6596

The scenario-only score ties by construction. In the additive score the scenario term cancels between two responses to the same scenario, so its ranking is exactly the response-only ranking. Its row decisions can still change. The tested relation features failed to improve within-scenario ranking on either split. The response-only score lost about 14.8 percentage points on the hard split. These two splits do not identify which cues caused that change. A dataset editor can use this matched diagnostic alongside row scores to check whether a model distinguishes plausible responses to the same request, while auditing label validity separately.

7 When the scenario must change the answer

Matching controls the scenario within each comparison. A benchmark can still reward a response-only score if the same responses seldom need different treatment across scenarios. For normalized response e, let $n_{e,0}$ and $n_{e,1}$ count its observed labels. Every deterministic response-only classifier makes at least $\sum_e \min(n_{e,0}, n_{e,1})$ row errors. Predicting the empirical majority for each response achieves that minimum on the given rows. That oracle uses the test labels and serves as an identifiability check, without estimating a deployable classifier's accuracy.

For rankings, let w_{ab} be the total macro weight of comparisons requiring response b to score strictly above a. Each mixed scenario has equal weight, split equally over its opposite-label pairs. With half error for ties, any response-only score incurs at least

$$\sum_{\{a,b\}} \min(w_{ab}, w_{ba})$$

macro error, where each unordered pair appears once. Ordering the responses either way incurs one directional weight. Tying them incurs half their sum. Each case costs at least the smaller weight. An additive scenario term cancels within each pair, so the same bound applies to additive scoring. Longer cycles can force error even when every term in the sum vanishes. Exact enumeration of weak orders verified the bound on 729 weighted three-response configurations, with equality in 675.

In the ordinary split, no normalized response carries both labels. In the hard split, six responses do, covering twelve rows and forcing at least six errors out of 2,800 for a response-only rule. Neither split contains an exact reversal of an observed response pair. The direct-reversal floor is therefore zero on both. This test exposes a small number of incompatible row labels. It cannot tell whether each conflict reflects context, a missing detail, or annotation error. It also cannot identify what caused the earlier ranking gap.

There are 67 ordinary and 66 hard rows whose normalized response appears in training. As an overlap sensitivity check, we dropped every test scenario containing any such response, leaving 511 ordinary and 642 hard scenarios. With the scorer fixed, response concordance remained 0.8037 and 0.6635, respectively. The gap persisted under the exact-overlap filter. Paraphrase and template

overlap remain untested. A prospective benchmark could use adjudicated pairs of scenarios in which the same two candidate responses trade places under a controlled change in permission or relevant facts. The resulting reversals would force a known error floor for any model that ignores the scenario.

Review status. Independent manuscript review and a language-model study remain open.

References

[1] M. J. Kusner, J. Loftus, C. Russell and R. Silva. Counterfactual Fairness. https://arxiv.org/abs/1703.06856.

[2] Thomas Zaslavsky. Signed graphs. Discrete Applied Mathematics 4, 47–74, 1982. https://davinci.fmph.uniba.sk/~matok1/diplomovka/sources/Zaslavsky-former.pdf.

[3] Richard M. Karp. Reducibility Among Combinatorial Problems. In *Complexity of Computer Computations*, 85–103, 1972. https://cgi.di.uoa.gr/~sgk/teaching/grad/handouts/karp.pdf.

[4] Dan Hendrycks et al. Aligning AI With Shared Human Values. *ICLR*, 2021. https://arxiv.org/abs/2008.02275.